

Solution Requirements

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Name	AI Study Buddy API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users must be able to register and log in securely. JWT authentication verifies users, while passwords are encrypted using bcryptjs.</i>	High
2	Authorization levels	<i>The system provides Student and Administrator roles. Students can access learning features, while administrators can manage</i>	High

		<i>users, monitor application activity, and maintain system integrity.</i>	
3	External interfaces	<i>The system connects with the Google Gemini AI API to generate summaries, flashcards, quizzes, and personalized study plans. MongoDB is used for data storage.</i>	<i>High</i>



4	Transactions processing	<i>The system accepts study materials from authenticated users, processes the content through the AI service, and returns generated learning resources through REST APIs.</i>	<i>High</i>
5	Reporting	<i>The system should display generated summaries, flashcards, quizzes, and personalized study plans. Users can view and manage previously generated learning resources.</i>	<i>Medium</i>
6	Business rules, etc.	<i>Users must authenticate before accessing AI-powered features. Uploaded study material is used as input for AI generation. Generated resources are associated with the respective user and can be stored for future retrieval.</i>	<i>High</i>

7	Compliance to laws or regulations	<i>The system should protect user information and credentials through authentication, authorization, and password encryption.</i>	Medium
8	Other	<i>The system must allow students to upload study materials and generate concise summaries, flashcards, multiple-choice quizzes, and personalized study plans.</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
------	--------------	-------------------------	-------------------------------------



1	Performance & Speed	<i>The system should process user requests efficiently and return AI-generated learning resources through REST APIs.</i>	<i>Responses should be processed within an acceptable time under normal usage.</i>
2	Scalability	<i>The modular RESTful architecture should support growth in users, study materials, and AI-generated resources.</i>	<i>System should support increasing users and data without major architectural changes.</i>
3	Security & Data Privacy	<i>User information and protected resources should be secured using JWT authentication, RBAC, and bcryptjs password encryption.</i>	<i>Only authenticated and authorized users can access protected resources.</i>

4	Reliability & Availability	<i>The backend should maintain reliable communication between the application, MongoDB, and AI service.</i>	<i>System should operate reliably when required services are available</i>
5	Usability & Accessibility	<i>Students should be able to easily upload study materials and access summaries, flashcards, quizzes, and study plans.</i>	<i>Learning features should be accessible through a clear and simple application flow.</i>